Ethical Theory and Moral Practice (2025) 28:569–583
https://doi.org/10.1007/s10677-025-10486-z

A Value-Free, Diversity-Sensitive Measure of Freedom

Nicolas Côté[1]

Accepted: 15 February 2025 / Published online: 24 February 2025
© The Author(s) 2025

Abstract
Numerous philosophers and economists have argued that the extent of one's freedom depends in part on how diverse one's options are. However, such efforts have met resistance from political philosophers who regard diversity-sensitive measures of freedom as objectionably value-laden. I present a new way of incorporating diversity considerations in the measure of freedom that is scrupulously value-neutral. It is also quite successful in capturing common intuitions about freedom and diversity: in particular, it is sensitive to the degree of dissimilarity between extremal and non-extremal options.

Keywords Diversity · Measure of freedom · Formal axiology · Measure theory · Social choice theory

1 Introduction

Intuitively, the extent of your freedom depends on how diverse your available actions are. Numerous philosophers and economists have supported this claim through persuasive examples (Raz 1986; Sen 1990; Klemish-Ahlert 1993; Suppes 1996; Rosenbaum 2000; Bossert et al. 1992; Pattanaik and Xu 2008). To take just one, a person looking to buy a car who enjoys a choice between ten differently-colored sedans enjoys greater freedom of choice than a person who only has a choice between ten same-colored sedans. Both prospective buyers may have the same number of options, but the first person gets to exercise choice over something which the second doesn't. Likewise, a person who can get from A to B by train or by bus enjoys more freedom of choice than someone who can get from A to B by a blue bus or by a red bus. In such examples, not only is it intuitive that greater diversity induces greater freedom, but, moreover, having a diversity of options is just one of those things that most of us *want*, insofar as we want freedom. In fact, in considering major life decisions, such as our choice of career, place of residence, where to school our children, many of us would prefer having a smaller number of highly dissimilar options to choose from rather than a larger number of samey options.

Intuitively then, if we want a measure of freedom that is extensionally adequate and captures what it is that people seem to care about in relation to their freedom, then it looks as though considerations of diversity must feature somewhere in the story. These sorts considerations have proved persuasive with moral philosophers interested in freedom as an indicator

✉ Nicolas Côté
Nicolas.Cote@glasgow.ac.uk

1 University of Glasgow, Glasgow, Scottland, UK

of well-being (e.g. Sen 1999). And yet, the attempt to sensitize the measure of freedom to diversity considerations has met significant resistance from political philosophers in the liberal tradition, on the grounds that such considerations are objectionably value-laden (e.g. Carter 1999; Hurka 2011; Bader 2018). For this reason, these theorists argue, we must exclude diversity considerations from the measure of freedom, because assessing how diverse options are requires us to make contentious value judgments on which differences between options matter.

In this paper, I defend a new way of measuring the diversity of individuals' freedoms which avoids the liberal critique. The core philosophical insight is that we can remove the need to take a stance on which differences between options matter by defining dissimilarity as an empirical magnitude, namely, the propensity of a pair of options to provoke strongly diverging evaluative responses on the part of randomly selected individuals. We can derive a cardinal and value-free measure of option dissimilarity. In turn, using this measure, I provide an axiomatic characterization of an attractive measure of freedom which is strictly increasing in both the number of options individuals are free to exercise, and in the diversity of these options. This measure itself doesn't break new mathematical ground: it is a special case of the multi-attribute measures of freedom characterized generally by Nehring and Puppe (2009). Nonetheless, it is an interesting special case because it satisfies a formal property that hasn't received any attention in the literature, but which provides one elegant solution to a common problem that diversity-sensitive measures of freedom encounter, namely: how to "reward" opportunity sets for having options that are increasingly similar to each other without simply ranking sets based on their most dissimilar pair of alternatives.

2 The Liberal Critique

What exactly is the problem with saying that you enjoy more freedom to the extent that the range of things you are free to do is increasingly diverse? The problem, say Garnett (2016) and Carter (1999, p. 121), is that assessments of option diversity ultimately rest on assessments of how dissimilar your options are from each other. Diversity is, in a sense, the aggregate level of mutual dissimilarity between your options (Nehring and Puppe 2009); a new freedom that opens up to you can only substantially increase the diversity of your exercisable freedoms if it is substantially different from the alternatives already on the table. Any assessment of dissimilarity must be objectionably value-laden, however, because it presupposes a point of view on which properties or quantitative dimensions *matter* for assessing dissimilarity. As Garnett (2016, p. 109) puts it: "it is presumably because we divide the world up into poets and baseball players, and not, say, into professions the names of which begin with different letters (poets and pitchers on one side, shortstops on the other) that we share" Crocker's (1980) judgment that we enjoy more freedom when we have a choice between being a baseball player or a poet than we do when we have a choice between being a pitcher or a shortstop.

Why is this a problem? For capability theorists like Sen (1999) and Nussbaum (2011), who wish to use freedom as the yardstick for human flourishing, you might think that the value-ladenness of a measure of freedom is a feature, not a bug. Judgments of well-being are, after all, moral judgments. If the point of the exercise is to build an informative and plausible measure of how well a person's life is going, then this measure of well-being simply *must* hook up with our judgments of what matters in life. I would argue that Conan's theory (from the 1982 film *Conan the Barbarian*) of what is best in life ("to crush your enemies, see them driven before you, and to hear the lamentations of their women"), while

informative, is implausible because it mistakes vices for virtues; I am here passing moral judgment on Conan's theory of the good. Similarly, if you want to measure flourishing in terms of freedom, then it had better be that the way you measure freedom is responsive to moral concerns. From that point of view, elevating certain evaluative perspectives over others in our measure of freedom (e.g. by privileging one way of dicing up the world into significant/meaningful properties) may be well to the point.

This is a seductive line of thought, but one which can be resisted on principled conceptual grounds. Freedom plays a fundamental role in many political theories, most notably the theory of liberalism. Garnett (2016) argues that if our conception of freedom is to serve the needs of liberal political philosophy, it should follow the liberal attitude of nonprejudiciality about what ends are worthwhile, for the familiar reasons that one's own value judgments may be wrong, that some disagreement about ends may be faultless, and that restrictions on people's freedom must be justifiable in terms acceptable to all, etc. But if judgments of diversity rest on contested value judgments about which properties matter, then we inevitably privilege some sectarian evaluative perspective over others. Carter (2018) further argues that political philosophy requires some basic value-free concepts to bridge moral and empirical discourses. These concepts may be concepts of values, or they may feature essentially in normative claims, but they must themselves be defined in purely descriptive terms to provide decidable truth-conditions for normative claims. Freedom is a good candidate basic concept because political philosophy in general, and liberal political philosophy in particular, "abounds with quantitative references to freedom" Carter (1995, p. 819). Freedom often serves to ground broader claims of justice – e.g., some institution is unjust because it carves excessive inroads into personal freedom. Broad claims of this nature have evaluative and descriptive content; in this case, the basic descriptive claim that grounds the overall claim of injustice is that some institution invades personal freedom (description) in a manner that is deemed excessive (evaluation). Assessing this claim for truth requires specifying empirical truth-conditions that can be tested against the world, and so a concept of freedom defined solely in descriptive, empirical parameters.

There are also moral grounds for wanting a value-free conception of freedom, rooted in precautionary reasoning. Even if you believe that we should "reward" opportunity sets for providing individuals with options that reflect *their* values or *their* interests, or interests that are universally shared, surely you should avoid rewarding opportunity sets for reflecting *your* values. That is parochial to a fault – it betrays the colonial and paternalistic mindset of one who thinks they know what's good for people better than they know themselves. Granted, the paternalist might be right, but they might be wrong, and they will be wrong all the more often they pontificate on how people should live (in the same way that the total number of false test results increases the more often you repeat the test). In light of this, it is good practice for moral and political philosophers to take precautionary steps against unwittingly colonizing those upon whom they would pass judgment. Giving ourselves fewer opportunities for "rigging" measures of freedom to reflect our prejudices looks like a reasonable precautionary step. These anti-colonialist, anti-paternalist considerations have all the more bite given that one of the primary contexts where freedom is invoked by philosophers to assess quality of life and of state institutions is that of global development – this is *the* context for which Amartya Sen and Martha Nussbaum developed their capability approach. And in this context, concerns over the dangers of colonialism have a unique salience.

In short, there are conceptual, political, and moral reasons to resist parochially value-laden conceptions of freedom. But at the same time, it's hard to shake the feeling that diversity really does matter to the assessment of freedom. If we wish to save the appearances without running afoul of liberal concerns, then we must show that option diversity reduces to some quality

whose measurement does not depend on (and therefore cannot be biased by) any evaluator's subjective value judgments.[1] The ideal here would be a reductive analysis of diversity in terms of an empirical magnitude like mass, velocity, or hue. The sections to follow will, accordingly, attempt a reductive analysis of option diversity in terms of the causal properties of options – specifically, their propensity to induce observable utility changes.

3 A Value-free Concept of Similarity

Our analysis begins with the same observation that prompted the liberal critiques: diversity supervenes on dissimilarity. What makes a collection of objects diverse is that the objects in the collection are dissimilar from each other, and the more dissimilar the members of the collection, the more diverse the collection. A reductive analysis of diversity therefore requires a reductive analysis of dissimilarity, and this is where our analysis truly takes off, with a few insights from economics.

It's a well-worn lesson from consumer choice theory that economic competition leads to product differentiation. If consumers cannot differentiate between the widgets sold by firm A and those sold by firm B, they will always prefer the cheaper widgets, and otherwise they will be indifferent between them. Firms which sell undifferentiated products can therefore only compete on price which will drive prices down to the marginal cost of production, and drive economic profits to zero. By contrast, if firms A changes its product line such that consumers perceive the new widgets to be different to the old ones (perhaps they have a different design, or different uses, or simply different branding), then consumers who prefer the new widgets will be willing to pay a premium for them. If, moreover, firm A is the sole producer of these special new widgets, it can exploit its market power to charge higher prices and earn higher profits. Thus economic competition leads every firm to strive to differentiate its products from its competitors: McDonald's sells Big Macs, Burger King sells Whoppers; Blue Bottle Coffee and Starbucks make their coffee using different beans and different recipes (and coffee drinkers, I'm told, can tell them apart).

This neat little story is often used to explain why market economies succeed where planned economies fail at satisfying the enormous diversity of human tastes. It also yields a natural analysis of what makes two goods similar to each other, namely, the fact that they are good *substitutes* for each other. Intuitively, two goods are good substitutes for each other if they can be used in the same way by consumers. More precisely, two goods are said to be *perfect substitutes* for one another if any price difference between the goods causes demand for the more expensive good to fall to 0 and shift to the cheaper good. By contrast, two goods are said to be *perfect complements* if demand for one good is completely invariant to the price of the other good. Goods that fall somewhere in between these two benchmarks are called *imperfect substitutes*.

Similarity and substitutability are not equivalent concepts, but in the context of consumer goods they are covariates. If two goods are similar in the respects that consumers care about, then they will behave like close substitutes. Why would you pay more for one widget than another if, for all you care about, they're the same? And if two goods are close substitutes, that can only be because they are similar in the respects that matter to consumers – for if consumers really found differences worth caring about between widgets A and B, they would be willing

[1] Or at least not their *moral* or *aesthetic* value judgments. If available measurement methods are subject to errors, different experimentalists may reach systematically different measurements if they weigh risks of error differently. This is a relatively inoffensive way in which judgments of diversity might be value-laden.

to pay more for the widget whose distinctive features they preferred. Substitutability therefore provides us with an empirically measurable proxy for similarity, and one which is value-free in the sense required by Carter and Garnett. Measuring how substitutable two goods are requires no assumptions about the respects in which goods can differ or the significance of those differences; it imposes no restrictions on the preferences consumers have or can have; it presupposes no moral, political, or aesthetic outlook; and the measurement cannot be rigged to suit any philosopher, politician, or civil servant's preferences. All the unobservable micro-level facts about all the ways in which goods can be different in ways that matter to someone are summarized without judgment or bias in the observable macro-level relationship between prices changes to one good and changes in total demand for another.

This analysis of similarity carries over from consumer goods to freedoms. Imagine a world in which individuals' opportunity sets were determined by the equilibrium outcome of a "market for freedoms," in which imaginary firms compete with each other to produce freedoms which can bought and sold (e.g. the freedom to eat an apple, to wear a pink shirt, to shout "fire" in a crowded theater, etc.); variably well resourced consumers purchase the bundles of freedoms they want, and the bundle of freedoms each individual ends up with when everyone has spent their money determines their opportunity set. Imagine too that the consumers interacting in this market make up a representative human population: anyone can express any preference, but the likelihood that a person randomly sampled from this population prefers, say, swimming in the Bay of Islands to fly-fishing in the Yukon, equals the likelihood that an actual human being randomly selected from the population here on Earth exhibits that same preference.[2] Some freedoms in this hypothetical market will prove to be very close substitutes for others, in that small changes in the price of one freedom will provoke large changes in demand for other freedoms; other freedoms will be more complementary. Since substitutability covaries with similarity, we can conclude that two options are similar to the extent that they would be good substitutes for one another in this hypothetical market.

Of course, no such hypothetical market exists, so we can't measure the substitutability of freedoms against this hypothetical market. But we can do something better. If two goods are perfect substitutes, then it follows that every individual must be indifferent between these goods – i.e. the average utility change induced by swapping one good for the other must be 0, because if even a single individual preferred one good to the other, they would be willing to pay a non-zero premium for it, and so demand for the preferred good will not fall to 0 for any price change. Conversely, the more complementary two goods are, the less indifferent between these goods individuals must be, on average. Indeed, the most general definition of substitutability is in terms of utility: two goods (or two freedoms) are increasingly/decreasingly close substitutes for one another the smaller/larger is the average utility change that would be induced by swapping one good for another. By the covariance of similarity and substitutability, it thus follows that the degree to which two options are similar to one another can be measured by the average difference in utility between these options which individuals chosen from a perfectly typical population would report. Equivalently,

[2] Is this not unduly limiting? The actual human population is not the only one that could have emerged, so isn't it parochial to set the actual distribution of preferences as the standard of reference? Maybe, but the present approach can be extended to account for such counterfactuals: imagine a two-step process in which we first randomly select a reference population (out of all the possible populations that could have emerged), and then randomly select an individual from that population. We now require that the likelihood that a person randomly sampled from our population of consumers prefers x to y equals the likelihood that an individual selected by this two-stage procedure exhibit this same preference. In practice this approach may not be so different from the one I have suggested, because we should assume that the actual human population is the kind of population most likely to emerge on an Earth-like planet.

since utilities are the numerical representation of preferences, which are in turn comparative value judgments instantiated in a disposition to choose, we might say that two options are increasingly similar to the extent that they invite increasingly close value judgments.

We can write this out precisely. Let X be the set of possible options, all mutually exclusive. What I have just argued is that the degree of similarity between two options is given by a function $\delta : X \times X \longrightarrow [0, +\infty)$ defined as $\delta(x, y) = \sum_{i=1}^{n} \frac{1}{n} |u_i(x) - u_i(y)|$, where $\{1, ..., n\}$ is a perfectly representative human population, and $u_i : X \longrightarrow [0, 1]$ is a cardinal utility function representing individual $i \in [1, n]$'s preferences over X. Again, the population is representative in the sense that the relative frequency of any given preference within this population is the objective chance that a randomly selected human being in the actual world exhibits this preference. An immediately appealing consequence of the proposal is that δ has all the properties of a metric:

1. Identity of indiscernibles: $\delta(x, y) = 0 \iff x = y$. The necessary part of this biconditional follows immediately from the fact that $u_i(x) = u_i(x)$: any option is a perfect substitute for itself; every individual is indifferent between one thing and itself; swapping one option for itself does not induce a change in utility. Sufficiency follows from the fact that any possible event has a non-zero probability of actually occurring, and since it is always possible that we might find someone who has a strict preference between x and y, for any $x \neq y$, and therefore the average difference in utility for x and y can never be zero in perfectly typical population. Two numerically distinct options are never perfect substitutes for each other, though they might be arbitrarily close to perfect substitutes.
2. Symmetry: $\delta(x, y) = \delta(y, x)$ for all x, y. This follows from the fact that $|u_i(x) - u_i(y)| = |u_i(y) - u_i(x)|$ for all i: the difference in your utility for x and your utility for y is the same as the difference in your utility for y and your utility for x. It is a conceptually nontrivial consequence of my definition that similarity satisfies symmetry, because folk concepts of similarity aren't in general symmetric: there's research dating back to Tversky (1977) suggesting that some people feel that North Korea is similar to China but not vice-versa.
3. Triangle inequality: $\delta(x, z) \leq \delta(x, y) + \delta(y, z)$ for all x, y, z. Another definitional consequence. The triangle inequality $|u_i(x) - u_i(z)| \leq |u_i(x) - u_i(y)| + |u_i(y) - u_i(z)|$ is true for all i, since $u_i(x), u_i(y)$, and $u_i(z)$ form three points on the real line, and therefore none of the line segments formed by connecting any two points can be longer than the sum of the other two line segments. Therefore $\sum_i^n |u_i(x) - u_i(z)| \leq \sum_i^n |u_i(x) - u_i(y)| + \sum_i^n |u_i(y) - u_i(z)|$, since whatever we add on the left with each i is no greater than whatever we add on the right. Multiplying both sides by $1/n$ gives us $\delta(x, z) \leq \delta(x, y) + \delta(y, z)$.

It should be noted that the definition of the similarity function assumes not only that individual utility functions are cardinal, but also interpersonally comparable, since we are averaging over different people's utilities. Cardinality might be defended as a modeling idealization. We know that individuals don't actually satisfy the axioms of expected utility theory, but we can pretend that they do. We impute to them the expectational utility function that most closely matches their actual pattern of preferences, and so long as the function we impute to them doesn't typically distort their actual preferences too much, we obtain a decent approximation of people's actual value judgments. But what about interpersonal comparability? Some philosophers believe utilities are interpersonally comparable (Adler 2011), but others deny it (Hausman 1995), and in general there is some reason to be skeptical. But in our particular context, we *can* make *some* interpersonal comparisons. We assumed that the set of possible alternatives X is finite. Therefore, we can induce a common scaling

across all utility functions by using Isbell's (Isbell 1959) zero–one rule: assign a utility of 1 to each individual's top-ranked alternative, and 0 to their bottom-ranked one. Under this choice of scaling, the utility assigned to an alternative tells us how much the individual cares about this alternative as a proportion of how much they care about their favorite alternative. And this is something we *can* compare across individuals.

The metric I proposed here bears a close resemblance to that proposed by Binder (2019, p. 54), which measures the dissimilarity of two options by the *number* of people (in a suitably chosen reference class) who express a strict preference for one over the other. The intuition behind Binder's metric and mine is the same: two options are dissimilar to the extent that they provoke divergent evaluations. The difference is that whereas my metric requires cardinal information about the intensity of individuals' preference for x over y, Binder's metric requires only ordinal information. Binder's metric is thus less informationally greedy than mine, but the cost of this economy is that it makes the resultant metric insensitive to difference among options that matter. If everyone *slightly* prefers x to y, but *strongly* prefers z to y, then y is presumably more dissimilar to z than to x. It must be if the intuition behind Binder's metric and mine is on the money. Yet whereas my metric correctly implies that $\delta(x, y) < \delta(z, y)$, Binder's incorrectly implies $\delta(x, y) = \delta(z, y)$. In the choice between my metric and Binder's we thus face a trade-off between extensional adequacy and informational economy. Thankfully, it does not matter for the results to follow which metric is used, so I will bracket this issue.

In either case, a startling upshot of the fact that similarity can be measured by a metric is that, despite having made no assumptions about the content of people's preferences, nor about the respects in which anyone cares that two options might differ, we can now infer exactly how many respects there are in which goods can differ in ways that matter to people. To see how this might work, think of an analogy with spatial reasoning: what is the difference between 2D space and 3D space? Quite simply, it is that in 2D space the maximum number of points that can all be equidistant from each other is 3 (the three vertices an equilateral triangle), while in 3D space the maximum is 4 (the four vertices of a regular tetrahedron). Moreover, if you know how all objects are spaced out relative to each other, you can immediately infer their relative positions: e.g. if a, b, c are all one unit distance from each other in the standard Euclidean metric, then their spatial coordinates (x_a, y_a), (x_b, y_b), (x_c, y_c) must be such that $(x_b - x_a)^2 + (y_b - y_a)^2 = 1$, and so on. What goes for physical space goes for similarity space: if we have a similarity metric, the number of dimensions along which options can be dissimilar is fixed by the maximum number of options that can all be equally dissimilar from each other. And since the metric tells us relative the relative distance in similarity between options, we can infer their relative positions in similarity space, and, therefore, how the options are distributed in this space.

There is a precise algorithm for doing all this. When relations between pairs of objects are modeled by a metric, the method of classic metric multidimensional scaling finds a set of points in a m-dimensional space such that each point represents one of the objects and the Euclidean distances between those points preserve the scores given by the metric fed to the algorithm (Cox and Cox 1978). The number of similarity dimensions m is unique, and the location of each option in similarity space (i.e. its coordinates) is unique up to distance-preserving transformation of the entire set of points (i.e. moving all the points by some set distance along a straight line, rotating all the points around the origin, or reflecting all the points through an axis or hyperplane). So in particular, if we feed this algorithm our metric δ, it will return a unique m and coordinates $d(x), d(y), \ldots$ in $\mathbb{R}^m$ for every element in X such that $||d(x) - d(y)|| = \delta(x, y)$ and these coordinates are unique up to distance-preserving transformations in $\mathbb{R}^m$.

4 A Diversity-sensitive Measure of Freedom

The above results permit an analysis of the diversity of an opportunity set in terms of how the elements of that set are dispersed over $\mathbb{R}^m$. There are different ways to do this, but I propose to measure the diversity of an opportunity set in terms of the volume of the similarity space that is enclosed by a set of bell curves that is associated to the opportunity set.

To see how this might work, imagine fixing an infinitely elastic sheet over an infinite plane, and then locally deforming the sheet by placing narrow pegs underneath it; locally, the sheet will reach a peak wherever we place a peg, then flatten out as we move away from it (though without ever becoming completely flat). A single peg thus turns our sheet into a 2-dimensional bell curve enclosing a unit volume of space. Now clearly, for every additional peg you place, you create more peaks, turning our sheet into a mountain landscape of bell curves which encloses an increasingly large volume. Likewise, as you drag all the pegs away from each other. This can be most clearly appreciated by considering how the volume changes where there are just two pegs to be placed: if you place your two pegs really close to each other, you barely deform the elastic sheet more than you would by placing a single peg. If some local region of the sheet is very close to some peak, then you can't create much extra volume under the sheet by creating a new peak in that exact region. The altitude of every point on the sheet is largely unchanged. In the limit, where you place the two pegs in the exact same spot, there's no difference between placing one and placing two: the volume stays the same. But if you place your second peg really really far away from your first peg, then you're placing this new peg in a region that is very nearly flat: by creating a new peak here you create a lot of extra volume underneath the sheet. In this image, pegs represent options, and their location on the plane represents their coordinates in $\mathbb{R}^m$. An opportunity set of pegs can therefore be represented by a unique mountain landscape, and the volume enclosed by that landscape represents the diversity of the set. A particularly diverse opportunity set is thus one whose mountain landscape encloses a particularly large volume.

This geometric picture can be stated precisely. Any bell curve can be defined by what is called a "Gaussian" function. In particular, the regular 1-dimensional bell curve which encloses a unit area is defined as the function $g : \mathbb{R} \to \mathbb{R}$ such that $g(t) = e^{-t^2}$. Having thus defined a 1-dimensional bell curve, we can define for each possible option an m-dimensional bell curve that reaches its peak at that option's location in similarity space and which encloses a unit volume. Let $Z = \{A \subseteq X | A \neq \varnothing\}$ be the set of all opportunity sets. Then, given $z \in X$ and $d(z) \in \mathbb{R}^m$, $g_z : \mathbb{R}^m \to \mathbb{R}$ is the m-dimensional bell curve $g_z(x) = g(\|x - d(z)\|)$. Finally, $D : Z \to \mathbb{R}$ is the diversity function defined by:

$$D(A) = \int_{\mathbb{R}^m} \max\left(g_{a_1}, \ldots, g_{a_k}\right).$$

D(A) is the volume enclosed by the mountain landscape of bell curves associated to the opportunity set $A = \{a_1, ..., a_k\}$.

Although I introduced D as a diversity function, I submit that it also supplies an attractive measure of freedom. Firstly, because it remains value neutral: it defines the degree of freedom offered by an opportunity set in terms of an objective measure of dispersion over a similarity space that is uniquely defined by empirically measurable relations of similarity. Facts about who enjoys more freedom than whom thus still reduce down to empirical facts about the average size of the utility shift induced by swapping one option for another.[3] Secondly, because it exhibits many desirable axiological properties.

[3] See also Suppes (1996), who measures dispersion using a measure of entropy.

(M) Monotonicity: if $A \subset B$ then $D(A) < D(B)$.

(Dis) Dispersion: for any set A and any option $x \in A$ such that $d(x)$ does not lie within the convex hull $\mathcal{K}$ of A, if $\mathcal{H}$ is a hyperplane that does not intersect the interior of $\mathcal{K}$, then $D(A \cup \{x\})$ strictly increases as $d(x)$ moves away from the convex hull of A along a line that is orthogonal to $\mathcal{H}$.

(C) Continuity: if x, y are two elements in A then $lim_{d(x) \longrightarrow d(y)} D(A) = D(A \setminus \{x\})$.

(II) Isometric Transformation Invariance: if B can be obtained from A by translation, rotation, or reflection, then $D(A) = D(B)$.

Theorem 4.1 *D satisfies M, Dis, C and II.*

The proof is in the appendix. From a formal point of view, D is a special case of Nehring and Puppe (2009) multi-parameter approach, so in that respect it does not break new conceptual ground. But it is an interesting special case nonetheless, because the above properties are appealing and, in the case of (Dis), unknown in the literature.

(M) says that each additional option you receive strictly increases the diversity of your choices, a basic desideratum on a measure of freedom. (Dis) rewards opportunity sets whose elements are more dispersed over $\mathbb{R}^m$. This condition has never been discussed in the literature, but I believe that it is a singularly attractive way of encoding a preference for diversity in our measure of freedom. Very roughly, it requires that as you "drag" one point from some set further and further away from the clump where the other points in the set are located, the set of points becomes more dispersed over $\mathbb{R}^m$. So the more dissimilar one option is from all of its alternatives, the more diverse your opportunity set is overall. And, conversely, the more similar one option is to your available alternatives, the less the addition of this option contributes to the diversity of your freedoms. Thus (Dis) captures in a natural way the intuition that diversity is about how mutually dissimilar all your options are – an intuition that has proven difficult to capture in a satisfactory manner (see van Hees 2004 for a discussion of various impossibility results).

(C) requires that an option which differs from another alternative only to a tiny extent can only make a tiny marginal contribution to the diversity of your opportunity set. Continuity is generally an appealing axiological property. To see why, consider the following example from Sugden (1992, p. 318): suppose you are offered a choice between one of three cans of the exact same beer on a flight, each different from the other only by their serial number and position on the tray. In this situation, says Sugden, you have *some* diversity of choice in this situation (choosing the can on the left is not *quite* the same as choosing the can on the right), but so vanishingly little one is almost tempted to say you only really have one option before you. Compare this to a situation where one of the three cans is swapped out for a bottle of wine.

(II), finally, states that if the options in A are as dissimilar, relative to each other, as the options in B are, relative to each other, then A and B must be equally diverse.[4] Or in other words, an option's absolute position in $\mathbb{R}^m$ has no bearing on how great a contribution it makes to the diversity of your options: only its relative position to available alternatives matters. This condition must be satisfied by any reasonable measure of diversity. Recall that the set of coordinates generated by the algorithm are only unique up to distance-preserving transformations, so a measure of diversity whose assessments were not invariant to such

[4] (II) entails the conditions (Klemish-Ahlert 1993) calls "indifference between shifted situations" and "indifference between reflected situations".

transformations would be completely pathological, as its assessments would change depending on an indifferent choice of coordinate system. Here's another way to think about it: if A can be obtained from B by translation, rotation, or reflection, this means that the elements of both sets have different intrinsic properties (since they are mapped to different locations in $\mathbb{R}^m$) but stand in the same similarity relations to each other. Diversity is a property of the relations in dissimilarity between options, not a property of the intrinsic features of options. Therefore changing the intrinsic properties while preserving the relational properties preserves the overall degree of diversity in the set. This is most evident when we consider two singleton sets. Any option x is as similar to itself as any other option y is similar to *it*self, so with no other basis for making judgments of diversity, we're left to conclude that $\{x\}$ and $\{y\}$ are necessarily equally (un)diverse.

What this last point reveals is that (II) implies a very familiar axiom on the measurement of freedom, the so-called Indifference between No-choice Situations (INS), which states that $D(\{x\}) = D(\{y\})$. (INS) is widely accepted in the literature on the measure of freedom, because the intuition behind it is simple and attractive: if two people both have no choice about what to do, then neither can be said to enjoy more freedom than the other. To illustrate, consider Able and Disable; if both were allowed to do as they pleased, Able would be able to choose between doing a, b, c, and d, while Disable would have a choice between doing a and b. As it is, however, both have no choice but to do a (e.g. $a =$ "lie still", and we have just put Able and Disable into a full-body brace that prevents all movement). In this scenario, where both are forced to do a without any choice about anything, both look equally unfree.

I am inclined to view it as proof of concept of my proposed measure of diversity that it entails (INS), but there are some who will instead view this as a *reductio* against my view. (INS) is notably inconsistent with the views of Steiner (1983); Carter (1999), and Kramer (2003), who argue that your freedom is not merely increasing in the number of actions you can pursue, it is also strictly *decreasing* in the number of actions you are physically prevented from doing. This view violates (INS) because it implies that Able truly is less free than Disable.[5] I find this conclusion rather odd. As the Able/Disable example shows, you can only prevent people from doing what they are able to do, thus a person who selectively disables themselves from being able to do those things which they are prevented from doing thereby minimizes the number of actions which they are prevented from carrying out, without reducing the number of actions they are free to do. On a view like Kramer's, this entails that their freedom has increased. Intuitions will differ, of course, but I confess to being rather pleased that D disallows the possibility of liberation through self-mutilation, by dint of satisfying (INS).[6]

[5] Hayek (1960, Ch 1), interestingly, is not committed to this result, for he denies that the extent of your freedom depends *at all* on what you can do: freedom, Hayek claims, is defined as the absence of coercion, and it is an abuse of the concept of freedom to suppose that it entails any ability to do anything we want (p. 65).

[6] I should note that Carter and Steiner (2022) insist that their view, but not Kramer's, also disallows the possibility of liberation through self-mutilation, because they, like Hayek and unlike Kramer, deny that the freedom to ϕ entails the ability to ϕ. Thus, prior to being forced to do a, Able and Disable are both free to do a, b, c, and d. I suppose this works as a response to the liberation-through-self-mutilation problem, but it jars with what they argue elsewhere. Steiner (1983, p. 74-5) and Carter (1999) argue that a person's degree of freedom is given by the ratio $\frac{|F_i|}{|F_i|+|U_i|}$, where F_i and U_i are the set of actions individual i is free and unfree, respectively, to perform. It's crucial to the appeal of this measure that the ratio is supposed to go down as $|U_i|$ goes up: Steiner writes that "to ignore [the actions which a person is prevented from doing] in estimating the extent of a person's liberty, is to misconstrue the object of such an exercise". But Carter (2022, p. 224) now insist that " the denominator [in the above ratio] is the same for every agent whose freedom we care to measure." But then, as they recognize, a person's degree of freedom does not depend in any way on the number of actions they are prevented from taking, because a person's freedom is simply proportional to the number of actions they are free to take.

But Carter (2004) thinks he has good reason to reject (INS). He argues that if you are forced to do x, and therefore your opportunity set is the singleton $\{x\}$, then it is right to say you enjoy no freedom *of choice*, but it is not right to say that you enjoy no freedom. As G.A. Cohen (1979, p. 9; 1986, p. 241-44) observes, you are free to do what you are forced to do. This undermines (INS), says Carter, because it can't be right to say you enjoy *no* freedom in no-choice situations, as (INS) claims, if there is something you are free to do. Kramer (2003, §5) makes the same point, alleging that if you accept (INS), then you can see no difference between the degree of freedom enjoyed by a dead person, which is nil, and the degree of freedom enjoyed a person who is forced to do something, which is more than nil. This objection rests on a confusion. (INS) does not assert that you enjoy "zero" freedom when you enjoy no choice; it merely asserts that you enjoy exactly as little freedom in that situation as anyone else who has no choice. If you want to stipulate that the degree of freedom they enjoy is greater than 0, to distinguish this from the degree of freedom enjoyed by the dead, which is 0, neither (INS) nor any of my axioms can stop you. Indeed, note that $D(\{x\}) = 1$ for any x and $D(\varnothing) = 0$, just as desired.

For all that, I don't mean to die on the hill that (INS) is a *sine qua non* condition on any measure of freedom. There can be no doubt that interference by others is a source of unfreedom that calls for special concern. This concern can motivate a rejection of (INS) – and therefore (II) – if one wishes to register it in the measure of freedom itself. One doesn't have to do this: one could preserve (II) and instead register the concern for non-interference in the measure of the *value* of freedom, as Pattanaik and Xu (2015) and I do in Côté (2020) when we suggest that one salient difference between mere inability and prevention is that the latter is a worse form of unfreedom. Or one can register this concern in a theory of rights, by imposing deontic constraints on deliberate interference in the lives of others. There are different ways of keeping the books, and I'm not sure that one must be preferred. More modestly, then, I simply want to insist that (INS) is a necessary condition on the measurement of freedom of choice, that is, the pure "opportunity aspect" of freedom. This, thankfully, is an aspect of freedom which everyone agrees matters.

In sum, the four conditions I have laid down all have real normative bite, and it speaks in favour of D that it satisfies them. Granted, D does not satisfy these four conditions uniquely, so the argument for using D rather than some alternative is weaker than it might be. That said, D's precise functional forms admits of an extremely satisfying axiological interpretation.

I argued in the previous section that similarity covaries with substitutability: if x and y are very similar, then they must be good substitutes for each other; having one is almost exactly like having the other. Moreover, if x and y are very similar, then the Gaussians representing them will reach their peaks at very nearby points, and in general they will have very similar values at every point in $\mathbb{R}^m$. In contrast, if x and y are very dissimilar, then their respective Gaussians will reach their peaks far away in each others' tails, mirroring the fact that x and y are very complementary. In a nutshell, the Gaussian associated with x distributes its density in a way that tracks how good a substitute x is for other possible options: where the density is high, x is a good substitute for any alternative whose Gaussian would peak at that point; where the density is low, x is a poor substitute. Put the pieces together and what we learn is that a very diverse opportunity set is a set that maximizes the space of possible options for which the options that are *actually* available are good substitutes.

This is not only a natural way of thinking about diversity – it also helps explain why diversity matters. If x and y are near-perfect substitutes, then having x gives one a sort of proxy access to y, even if you don't have y, so having more diverse options is a way of having access to a greater number of options – a way of having more options without actually having more options. This gives us a way of motivating the concern for diversity in a way that should

be acceptable even to philosophers like Ian Carter, Hillel Steiner, and others who believe that the measure of freedom should be based on purely on how many freedoms you have or don't have. At the same time, this explanation supports the usefulness of freedom as an indicator of human flourishing. A highly diverse opportunity set must capable of satisfying a broad range of desires. It also provides a more meaningful context of choice: options which are highly dissimilar to each other are options which invite sharply contrasting value judgments; to choose just one option out of a diverse set is therefore to seriously engage one's evaluative faculties. This is something which individuals have reason to care about, and which the state has good reason to promote: individuals because they are more likely to find their own preferences satisfied and it enables them to exert meaningful authorship over their lives, and the State because the promotion of freedom can serve as a reliable way of promoting individuals' interests without taking controversial stands on what makes life go well.

So far, the argument for D has proceeded at a purely theoretical level. But how well does the model drive off the lot? It clearly gets the easy cases right: if you and I both enjoy two options, but mine are more dissimilar than yours, then I necessarily enjoy more freedom than you do. That's a straightforward consequence of (Dis) + (II). A bigger challenge is whether it handles "intermediary" options well. This is a well-known problem for other theories of diversity-measurement, such as Bervoets and Gravel (2007), which orders opportunity sets by the dissimilarity of their most dissimilar pair of options. Happily, D shines in this area, as the following example will show.

Electoral choice: election season is underway, and on day 1 there are two political candidates you can vote for, one on the extreme right (x) and another on the extreme left (y). Let's suppose that the assignment of coordinates to these options returns coordinates of -100 and +100 on the 1-dimensional left-right spectrum. On day 2 a third candidate enters the race. Let us consider two possible cases of their political alignment.

- case 1: the new candidate (z) is a perfect centrist, sitting at coordinate 0.
- case 2: the new candidate (w) is another lefty, sitting at coordinate -70

In each case, how does your overall degree of freedom change from day 1 to day 2?

On Bervoets and Gravel's view, the answer is, in both cases: not at all. The opportunity sets $\{x, y\}$, $\{x, y, z\}$, and $\{x, y, w\}$ offer exactly the same degree of freedom, because they share each other's most dissimilar pair of alternatives. In contrast, not only does D correctly imply that the addition of the intermediary candidate increase your electoral freedom (a trivial consequence of M), it implies that your electoral freedom is maximized in case 1: $D(\{x, y\}) < D(\{x, w, z\}) < D(\{x, y, z\})$. This because if you start with two bell curves (g_x and g_y) and you add a third in between them, the area of overlap between that middle bell curve and either of the outer two bell curves increases exponentially as the peak of the middle bell curve approaches the peak of the outer bell curve, thus the total area of overlap is minimized when the middle bell curve reaches its peak exactly at the midpoint of the distance between the two outer peaks. This is intuitively the correct conclusion: you have more electoral freedom of choice when you have a choice between a leftist, a centrist, and a right-winger than when you have a choice between two lefties and a right-winger. In the same way that have more marital freedom if you have a choice between marrying a vegan, an omnivore, and a carnivore than if you have a choice between marrying a vegan, a vegetarian, and a carnivore. The configuration of points in case 1 maximizes the mutual degree of dissimilarity of your options – it maximizes the range of political positions for

which you have a close substitute and in all likelihood, it also maximizes the number of voters who will find at least one candidate that they find acceptable.

Finally, it is also to be noted that if you substitute x and y for candidates $x + \epsilon$ and $y - \epsilon$, who are infinitesimally closer to 0 than x and y, then Bervoets and Gravel's measure of freedom implies that $\{x, y\}$ offers strictly more freedom than $\{x + \epsilon, z, y - \epsilon\}$, while D correctly implies that, for ϵ not too large, the reverse holds, as $D(\{x + \epsilon, z, y - \epsilon\}) \approx D(\{x, y, z\}) \approx 3 > D(\{x, y\}) \approx 2$. And this seems right: your options remain overall more diverse on day 2 than on day 1 even if x and y both infinitesimally moderate their positions. The diversity to be gained from additional options can swamp the diversity to be lost from making all your options slightly more similar. Looking at all these cases, it appears that D handles the road pretty well.

5 Conclusion

This paper has argued that there exists a measure of freedom which adequately captures our concerns both for having *many* options to choose from, and for having a *diversity* of options to choose from. This measure, D, satisfies the liberal requirement of value-freeness, it rewards every increase in the number of freedoms a person can exercise and in their degree of mutual dissimilarity, it rewards more those freedoms which are very dissimilar to their alternatives while rewarding almost not at all those freedoms which are nearly identical to their alternatives, it is insensitive to any information other than the number and degree of mutual dissimilarity of our options (which would be superfluous for a measure of diversity), and it tests well against the usual cases that pose problems for other measures.

Appendix

Proof 1. I first show that D satisfies monotonicity. Let A, B be arbitrary sets such that $A \subset B$. By definition, for any $b_i \in B \backslash A$, the function g_{b_i} reaches its maximum value of 1 at $x = d(b_i)$, while for every other $a_i \in A$, g_{a_i} has a value of strictly less than 1 at $x = d(b_i)$. Therefore there is a collection of open intervals in $\mathbb{R}^m$ such that $max(g_{a_1}, ..., g_{a_n}, g_{b_1}, ..., g_{b_m})$ is greater over those intervals than $max(g_{a_1}, ..., g_{a_n})$. These open intervals will be the neighbourhood of points around each point $d(b_i)$ that are strictly closer to $d(b_i)$ than any other element of B, which are obviously open intervals since B is a finite set of points. Further, there is no point in $\mathbb{R}^m$ over which $max(g_{a_1}, ..., g_{a_n}, g_{b_1}, ..., g_{b_m})$ is strictly lesser than $max(g_{a_1}, ..., g_{a_n})$. Therefore $D(B) > D(A)$.

2. Showing that D satisfies dispersion is harder work. Let's start by considering a set A of points $d(a_1), ..., d(a_m) \in \mathbb{R}^m$. Let $\mathcal{K}$ be the convex hull of A and assume that $d(a_1)$ does not lie in the interior of $\mathcal{K}$. Let $\mathcal{H} \subset \mathbb{R}^m$ be a hyperplane which contains $d(a_1)$ and which does not intersect the interior of $\mathcal{K}$. Let $\mathbf{v}$ be a unit length vector orthogonal to $\mathcal{H}$.

Let us now "pull" $d(a_1)$ away from $\mathcal{K}$ while keeping $d(a_2), ..., d(a_m)$ fixed. More precisely, for $s \geq 0$, we consider the family $d(a_1, s) := d(a_1) + s\mathbf{v}$. We let $g_s(x) = max_i(e^{-|x-d(a_i,s)|^2})$ and we let $D(s) = \int_{\mathbb{R}} g_s$. Note that g_s is simply an abbreviated notation for the integrand of my diversity function D, and accordingly $D(s)$ denotes the diversity of the set of options $a_1, ..., a_m$ as we vary the position of $d(a_1)$. What we want to show is that as we "pull" $d(a_1)$ away from all the other points – i.e. as we make a_1 increasingly dissimilar from every other alternative in A – the diversity of A increases.

More precisely, we wish to prove that $D(s)$ is nondecreasing. To prove this, we begin by taking an arbitrary point $\mathbf{b} \in \mathcal{H}$. We consider the line $l(t) = t\mathbf{v} + \mathbf{b}$. We will now study the restriction of g_s to the line l.

By applying an affine transformation and choosing an appropriate orthogonal basis we may as well assume that $\mathbf{v} = e_1$ and that $\mathcal{H} = \text{span}\{e_2, ..., e_n\}$. Then $g_{s|l} = max_{i=1}^{m}(C_i e^{-(t-(d(a_i))_1(s))^2})$, where $C_i = e^{-(d(a_i)_2)^2}...e^{-(d(a_i)_n)^2}$. Note that $(d(a_1)_1(s) = s$ while for $j = 2, ..., m$, we have $(d(a_j))_1(s) = (d(a_i))_1$.

Lemma 5.1 *If $d(a_i)_1 \leq 0$ for $i = 2, ..., m$, then $\int_l g_{s|l}$ is nondecreasing in s, for $s \geq 0$.*

Proof By translation, we may as well assume $s = 0$.

It's enough to show that the set of $t \in \mathbb{R}$ on which e^{-t^2} is greater or equal to $Ce^{-(t-\alpha)^2}$ forms an interval of the form $[d, \infty)$. – if $\alpha = 0$, then this is clear. So assume $\alpha < 0$.

We consider the quotient $e^{-t^2}/(Ce^{-(t-\alpha)^2}) = e^{a^{2-2t\alpha}}/C$. We ask when this is ≥ 1. Taking logs, this is the same as asking when $-logC + (-2t\alpha + \alpha^2)$ is ≥ 0. i.e. $t \geq (-\alpha^2 + logC)/2|\alpha|)$. The right-hand term of this inequality is a real number d, thus $t \in [d, \infty)$. □

Corollary 5.1.1 *$D(s) = \int_{\mathbb{R}^{n-1}}(\int_{l_\mathbf{b}} g(s)|l_\mathbf{b})$ is nondecreasing.*

Proof Since the integrand is nondecreasing, the integral is nondecreasing. □

3. D is trivially continuous. Gaussian functions are continuous, the max function is continuous, and the integral function is continuous. Any function which can be decomposed into continuous functions of other continuous functions is itself continuous.

4. Trivial. It is a definitional consequence of D being an integral over $\mathbb{R}^n$ that it is invariant to isometric transformations. □

Acknowledgements My many thanks to Campbell Brown and Alex Voorhoeve, whose advice has been invaluable in writing this article, as well as to reviewers from Economics and Philosophy and Ethical Theory and Moral Practice for their insightful and probing comments, which have helped me to significantly improve my manuscript. My thanks also to Philip Pettit, with whom I had many engaging conversation which were instrumental to the completion of this paper. Finally, special thanks to Laurent Côté and Lisa Sauermann, who were indispensable to me in working out the characterization of the diversity function and proving that it satisfied the desired conditions.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Adler M (2011) Well-being and fair distribution: beyond cost-benefit analysis. Oxford University Press, Oxford

Bader R (2018) Moralizing liberty. Oxford Stud Pol Philos 4:141–166

Bervoets S, Gravel N (2007) Appraising diversity with an ordinal notion of similarity: an axiomatic approach. Math Soc Sci 53:259–273

Binder C (2019) Agency, freedom and choice. Springer

Bossert W, Pattanaik PK, Yongsheng X (1992) Similarity of options and the measurement of diversity. J Theor Pol 15:405–21

Carter I (2018) Value-freeness and value-neutrality in the analysis of political concepts. Oxford Studies in Political Philosophy, Volume 1. Ed. by David Sobel. Oxford: Oxford University Press
Carter I (1995) The independent value of freedom. Ethics 105(4):819–845
Carter I (1999) A measure of freedom. Oxford University Press, Oxford
Carter I (2004) Choice, freedom, and freedom of choice. Soc Choice Welf 22:61–81
Carter I, Steiner H (2022) Freedom without trimmings: the perils of trivalence. Without trimmings: the legal, moral, and political philosophy of matthew kramer. Oxford University Press
Cohen GA (1979) Capitalism, Freedom, and the Proletariat. The idea of freedom: essays in honor of isiah berlin. Ed. by A. Ryan. London: Oxford University Press
Cohen GA (1986) History, labour and freedom. Clarendon Press, Oxford
Côté N (2020) Weakness of will and the measurement of freedom. Ethics 130(3):384–414
Cox M, Cox T (1978) Multidimensional Scaling. Handbook of data visualization. Ed. by Springer Handbooks Comp. Statistics. Berlin, Heidelberg: Springer
Crocker L (1980) Positive liberty. Nijhoff, London
Garnett M (2016) Value neutrality and the ranking of opportunity sets. Econ Philos 16:99–119
Hausman D (1995) The impossibility of interpersonal utility comparisons. Mind 104(415):473–490
Hayek F (1960) The constitution of liberty. The University of Chicago Press, London
Hurka T (2011) Drawing morals: essays in ethical theory. Oxford University Press, Oxford
Isbell JR (1959) Absolute Games. Contributions to the theory of games, Vol. IV. Ed. by A. W. Tucker and R. D. Luce. Princeton: Princeton University Press, pp 357-96
Klemish-Ahlert M (1993) Freedom of choice. a comparison of different rankings of opportunity Sets. Soc Choice Welf 10:289–173
Kramer M (2003) The quality of freedom. Oxford University Press, Oxford
Nehring K, Puppe C (2009) Diversity. The handbook of rational and social choice. Ed. by Paul Anand, Prasanta K. Pattanaik, and Clemens Puppe. Oxford: Oxford University Press, pp 298-318
Nussbaum M (2011) Creating Capabilities: the human development approach. Cambridge. Cambridge, MA: Harvard University Press
Pattanaik PK, Xu Y (2008) Ordinal distance, dominance, and the mea- surement of diversity. Rational Choice and Social Welfare. Ed. by Prasanta K. Pattanaik et al. Berlin: Springer, pp 259-269
Pattanaik, PK, Xu Y (2015) Freedom and its Value. The Oxford Handbook of Value Theory. Ed. by I. Hirose and J. Olson. Oxford: Oxford University Press
Raz J (1986) The morality of freedom. Oxford University Press, Oxford
Rosenbaum EF (2000) On measuring freedom. J Theor Pol 12:205–77
Sen A (1999) Development as freedom. Knopf Press
Sen A (1990) Welfare, Freedom, and Social Choice: A Reply. Recherches Economiques de Louvain 56:451–85
Steiner H (1983) How free: computing personal liberty. Royal Institute of Philosophy Lectures, pp 73-89
Sugden R (1982) The metric of opportunity. Econ Philos 14:307–337
Sugden R (1992) The metric of opportunity. Econ Philos 14:307–337
Suppes P (1996) The nature and measurement of freedom. Social Choice and Welfare 13(2):183–200
Tversky A (1977) Features of similarity. Psychol Rev 84(4):327–352
van Hees M (2004) Freedom of choice and diversity of options: some difficulties. Soc Choice Welf 22:253–266

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.